

Feasibility Memo Outline

AI-Assisted Triage: Data Exploration (Week 5 Interim Submission)

Preliminary Verdict

The dataset appears suitable for developing an AI-assisted triage model, provided that identified data quality issues are addressed and the model's applicability outside the original hospital setting is carefully considered.

Dataset Summary

- The dataset contains **55,121 patient records** and **225 clinical features**.
 - The target variable is the **Emergency Severity Index (ESI)**, which classifies patient acuity on a scale from **1 (most urgent)** to **5 (least urgent)**.
 - Variables include:
 - Demographic information (age, gender, race, ethnicity)
 - Vital signs (heart rate, blood pressure, respiratory rate, oxygen saturation, temperature, glucose)
 - Administrative information
 - Approximately 200 binary chief complaint (**cc_**) variables.
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Top Three Data Quality Concerns

1. Sparse Chief Complaint Variables

Many chief complaint (**cc_**) variables occur in fewer than 0.5% of patients. Their low prevalence may contribute little predictive value individually and may introduce unnecessary noise into a predictive model.

2. Outliers and Data Quality

Several vital signs contain statistical outliers, while a small number of clinically impossible values were identified and should be treated as missing or corrected before modelling.

3. Limited Generalisability

The dataset was collected from a single United States emergency department. Differences in patient demographics, healthcare systems, and disease prevalence may limit direct application to Caribbean emergency departments.

Top Three Reasons to Proceed

1. Large Dataset

The dataset contains over 55,000 patient encounters, providing a strong foundation for machine learning.

2. Clinically Relevant Variables

Important predictors such as age, vital signs, and chief complaints are available and are directly related to emergency triage decisions.

3. Good Overall Data Completeness

The structured variables examined during exploratory analysis showed no missing values after the cleaning process, indicating good overall data quality for modelling.

Caveats

- Variables representing outcomes after triage (for example, **disposition**) must be excluded to prevent data leakage.
- Model fairness should be evaluated across demographic groups to reduce the risk of bias.
- Additional validation will be required before applying the model in Caribbean healthcare settings.

Preliminary Top-10 Feature Shortlist

The following features are expected to be among the strongest predictors of ESI based on clinical relevance and exploratory analysis:

1. Age
2. Heart rate
3. Respiratory rate
4. Oxygen saturation
5. Systolic blood pressure
6. Temperature
7. Blood glucose
8. Chest pain
9. Shortness of breath

10. Arrival mode

Next Steps

- Complete data cleaning and preprocessing.
- Confirm feature importance through statistical analysis.
- Develop and evaluate baseline machine learning models in Week 6.
- Compare model performance and assess fairness before recommending deployment.